Object Manipulation in Virtual Reality using Hands: Systems and Implementation

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***Abstract*— We propose a system of haptic rendering and interactive simulation which enables elaborate hand-object manipulation in virtual reality (VR). We unify the visual-inertial skeletal tracking (VIST)-based hand-tracking system, passive midpoint integrator (PMI)-based interactive simulation, and tactile feedback device, accomplishing a novel, stable and robust hand simulator in VR. The hand-tracking part employs VIST algorithm, tightly-coupled sensor fusion of vision and inertial sensors, which is robust against occlusion, interference, and contact. PMI-based simulation approximately preserves passivity and non-iteratively calculates each time-step, so that it enforces stability and real-time immediacy. We also verify this system implementing in two different scenarios: peg-in-hole and ball manipulation.**

I. INTRODUCTION

As the COVID-19 virus has spread rampantly, many research groups and companies are developing metaverse systems to enable class, meeting, conference, shopping and other everyday life tasks in a contact-free way. Current existing metaverse systems (e.g. VRChat [5], ZEPETO [6], Gather [1], etc.) are suitable for communication or simple tasks like holding and releasing objects in on/off way but, it is still impossible to elaborately manipulate virtual objects with existing commercial systems. According to this circumstance, a number of studies for realistic haptic rendering and interactive simulation combining hand-tracking technology have been proposed [10], [8]. In line with this trend, we propose another new system for haptic rendering and interactive simulation using a haptic glove.

We adopt the idea of passive midpoint integrator (PMI) [10] to preserve passivity and enforce stability of simulation. The concept of virtual proxy is applied to connect haptic glove with virtual world by virtual coupling, the way preserving passivity [11], [7].

Meanwhile, dexterous control of the hand is mandatory to handle the object elaborately, therefore tracking and simulating the human hand is essential for delicate and complex object manipulation in a virtual environment. However, accurate and robust tracking of the human hand is a very difficult problem because the structure of the human hand has high degree-of-freedom in a compact space. There have been several solutions for hand-tracking technology and visual-inertial skeletal tracking (VIST) method [12] is the state-of-the-art. Vision-based hand-tracking methods (e.g. Google MediaPipe [2]) are vulnerable to occlusion. Inertial measurement units (IMUs) with compasses-based hand-tracking methods (e.g. Prime X [3]) are vulnerable to magnetic interference. Soft stretch sensor-based hand-tracking methods (e.g. Sense Glove [4]) are vulnerable to contact and not durable due to hysteresis. We adopt VIST system for hand-tracking because this system is robust against occlusion, magnetic interference, and contact.

In this paper, we propose the unified system of VIST-based hand-tracking, PMI-based interactive simulation, and haptic rendering with two modalities - visual and vibrotactile feedback. The VIST system and the haptic feedback device are combined into a simple, wearable glove form.

The rest of this paper is composed of system design, scenarios, and a conclusion. In Section II, the layout of the hardware setup is depicted, and each major components of the whole system are described in detail. In Section III, two scenarios for the implementation of our system are described with snapshots. Lastly, Section IV concludes and summarizes.

II. SYSTEM DESIGN

A. Hardware Setup

The overall experimental hardware setup is composed of three parts - a monitor, an RGB stereo camera (ZED mini), and a haptic glove. The monitor, which also can be alternated with a head-mounted-display (HMD), provides an artificial visual rendering of VR environment. The RGB stereo camera is used to obtain visual information for VIST-based hand-tracking system. The haptic glove tracks the motion of hand with three fingers (thumb, index finger, and middle finger) and gives vibrotactile feedback. The haptic glove has a double-layered structure - inner glove and outer glove. Inside the inner glove, piezoelectric actuators are attached to each fingertips. 7 IMUs sensing 6-axes information are attached between the inner and the outer gloves, and circular fabric markers with 4 colors (red, blue, green, yellow) are attached to the outside of the haptic glove.

B. VIST-based Hand-tracking System

VIST-based hand-tracking system achieves highly accurate hand and finger tracking performance with 10 mm-

This work was supported by Institute of Information & communications Technology Planning & Evaluation(IITP) grant funded by the Korea government(MSIT) (No.2021-0-00896, Development of 3D collaborative teleoperation technology for unstructured work in harsh environments), the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No.NRF-2016R1A5A1938472), and the Industrial Strategic Technology Development Program grant funded by the Korea government(MOTIE) (No.20001045).

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Fig. 1. Hardware setup consist of monitor (red outline), RGB stereo camera (green outline), and haptic glove (blue outline).

scale mean error. VIST algorithm complementarily fuses visual and inertial sensors in tightly-coupled manner, in other words, IMU information is used for correspondence search of vision markers, and vision information corrects uncertainty of IMU sensors data. As a result, it overcomes both shortcomings of vision-based hand-tracking, which is vulnerable to vision occlusion, and IMU with compasses-based hand-tracking, which is vulnerable to magnetic interference. Furthermore, VIST algorithm does not use any soft sensors or compasses therefore it can follow hand motion even if part of the hand is occluded or out of the camera's field of view (FOV); or, the hand touches or grasps the object; or, electronic devices or magnets are near the hand.

C. Simulation

The VIST algorithm calculates the human hand configuration at a rate of about 85 Hz, and calculated human hand configuration is connected to the virtual proxy of the human hand by the virtual coupling, which is the combination of virtual spring and damper. We exploited PMI for the update rule of dynamics to approximately preserve passivity of the simulation, and the update rate varies between 50-200 Hz according to the computational load. We check regularly spread points on the surface of rigid bodies to detect collision among the rigid bodies. Contact force is calculated solving nonlinear complementarity problem (NCP) induced from Coulomb's friction model with projected Gauss-Seidel (PGS) method, and the details of NCP PGS algorithm are explained by Horak and Trinkle [9].

D. Tactile Haptic Feedback

The haptic glove serves the role of a hand-tracker along with a haptic device. The haptic glove is equipped with three haptic modules located on each finger - thumb, index finger, middle finger - to give vibrotactile feedback when the virtual proxy of the hand comes in contact with other



Fig. 2. VIST-based hand-tracking glove composed of two layers of gloves, IMU sensors, and fabric markers.

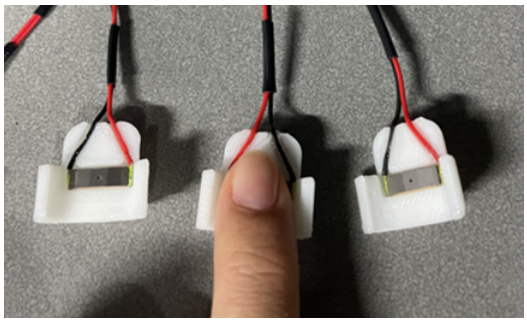


Fig. 3. Vibrotactile modules, which are attached to the fingertips of the inner glove with their compact size (12×4 mm): piezoelectric actuators are used.

rigid bodies. The haptic modules are attached to the inside of the inner glove with compact size (12×4 mm). These modules can render delicate differences in vibration due to the use of piezoelectric actuators, which have advantages in delicacy and immediacy compared to eccentric rotating mass (ERM) motor and linear resonant actuator (LRA). The intensity of the vibrotactile feedback is decided by the contact force calculated through NCP PGS method referred to in *Simulation* part, and it increases as the contact force increases.

III. SCENARIOS

Here we perform two illustrative scenarios of peg-in-hole task and ball manipulation.

A. Peg-in-hole

Fig. 4 shows the serial process of performing peg-in-hole scenario. The user grips the peg, and inserts it into the hole.

B. Ball Manipulation

Fig. 5 shows the snapshots of performing ball manipulation scenario. The user rolls, grips, and performs various in-hand manipulation with a ball.

IV. CONCLUSION

In this paper, we proposed the unified system of VIST-based hand-tracking, PMI-based dynamical simulation, and tactile haptic feedback. Robust and stable performance of

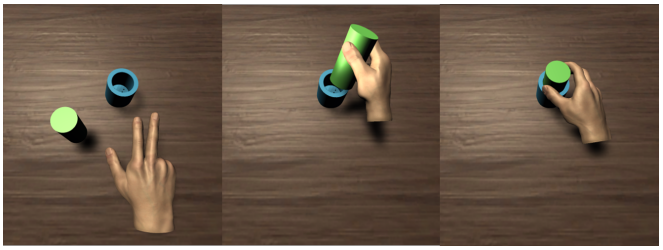


Fig. 4. Snapshots of the scenario that the user tries to insert the peg into the hole (i.e. peg-in-hole).

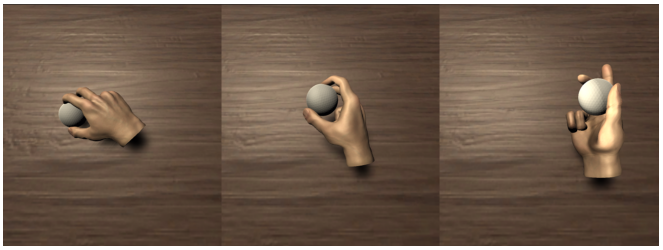


Fig. 5. Snapshots of the scenario that the user rolls and holds a ball with various kinds of grip.

VIST-based hand-tracking system allows to manipulate virtual objects elaborately even against occlusion, magnetic interference, and contact. PMI and virtual coupling techniques assure the stability of dynamical simulation, and NCP PGS method iterates fast enough for real-time contact simulation. Vibrotactile feedback modules have a compact design to be combined into hand-tracker glove, and they can render delicate vibration immediately using piezoelectric actuators. We verified the system performance with two scenarios, one is peg-in-hole and the other is ball manipulation. This system can be used in metaverse platforms, enabling elaborate object manipulation, and, by extension, many applications are possible (e.g. VR vocational training, human-robot interaction, etc.).

REFERENCES

[1] Gather. www.gather.town.
[2] Google mediapipe. google.github.io/mediapipe.
[3] Prime x. www.manus-meta.com.
[4] Senseglove. www.senseglove.com.
[5] Vrchat. hello.vrchat.com.
[6] Zepeto. zepeto.me.
[7] R. J. Adams and B. Hannaford. Stable haptic interaction with virtual environments. *IEEE Transactions on robotics and Automation*, 15(3):465–474, 1999.
[8] C. Garre, F. Hernández, A. Gracia, and M. A. Otaduy. Interactive simulation of a deformable hand for haptic rendering. In *2011 IEEE World Haptics Conference*, pages 239–244. IEEE, 2011.
[9] P. C. Horak and J. C. Trinkle. On the similarities and differences among contact models in robot simulation. *IEEE Robotics and Automation Letters*, 4(2):493–499, 2019.
[10] M. Kim, Y. Lee, Y. Lee, and D. Lee. Haptic rendering and interactive simulation using passive midpoint integration. *The International Journal of Robotics Research*, 36(12):1341–1362, 2017.
[11] D. Lee, M. Kim, and T. Qiu. Passive haptic rendering and control of lagrangian virtual proxy. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 64–69. IEEE, 2012.
[12] Y. Lee, W. Do, H. Yoon, J. Heo, W. Lee, and D. Lee. Visual-inertial hand motion tracking with robustness against occlusion, interference, and contact. *Science Robotics*, 6(58):eabe1315, 2021.